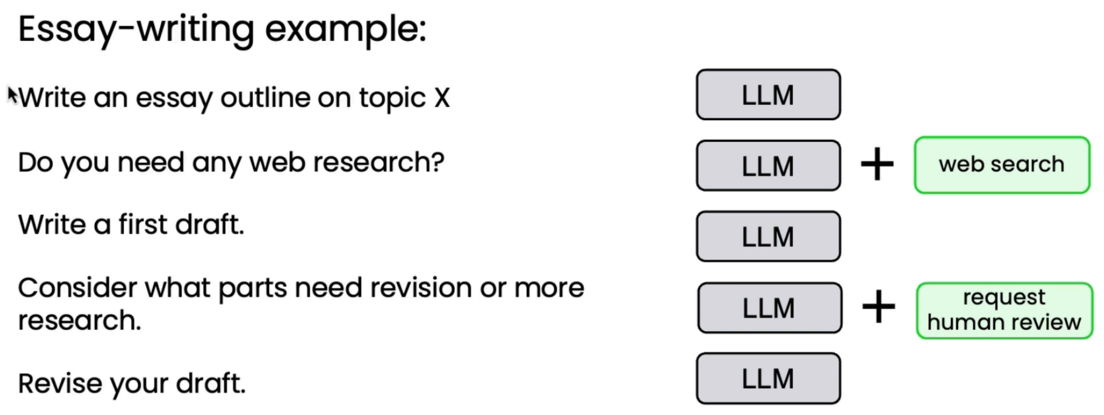
Agentic AI

Non-agentic workflow (zero-shot)

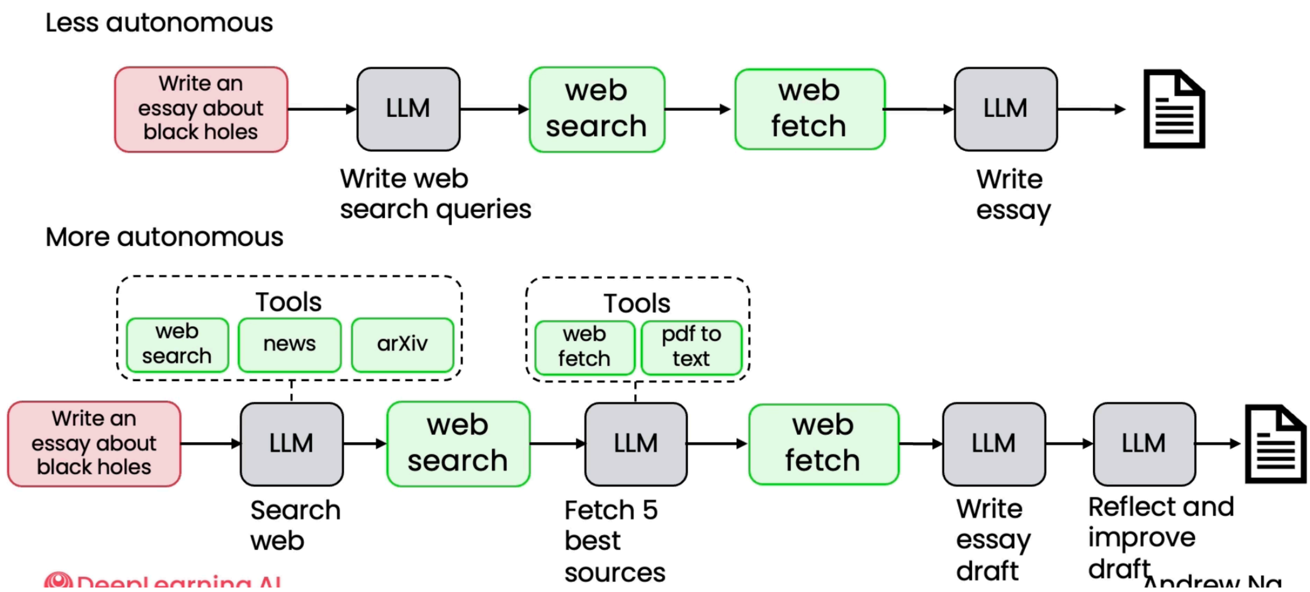
An agentic AI workflow is a process where an LLM-based app executes multiple steps to complete the task.



Introduction

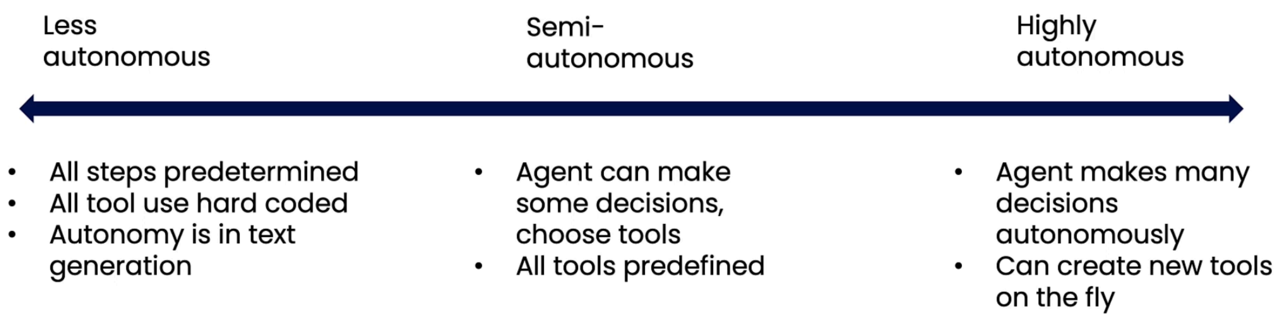
"Rather than arguing over which work to include or exclude as being a true agent, we can acknowledge that there are different degrees to which systems can be agentic."
Andrew Ng

Degrees of Autonomy

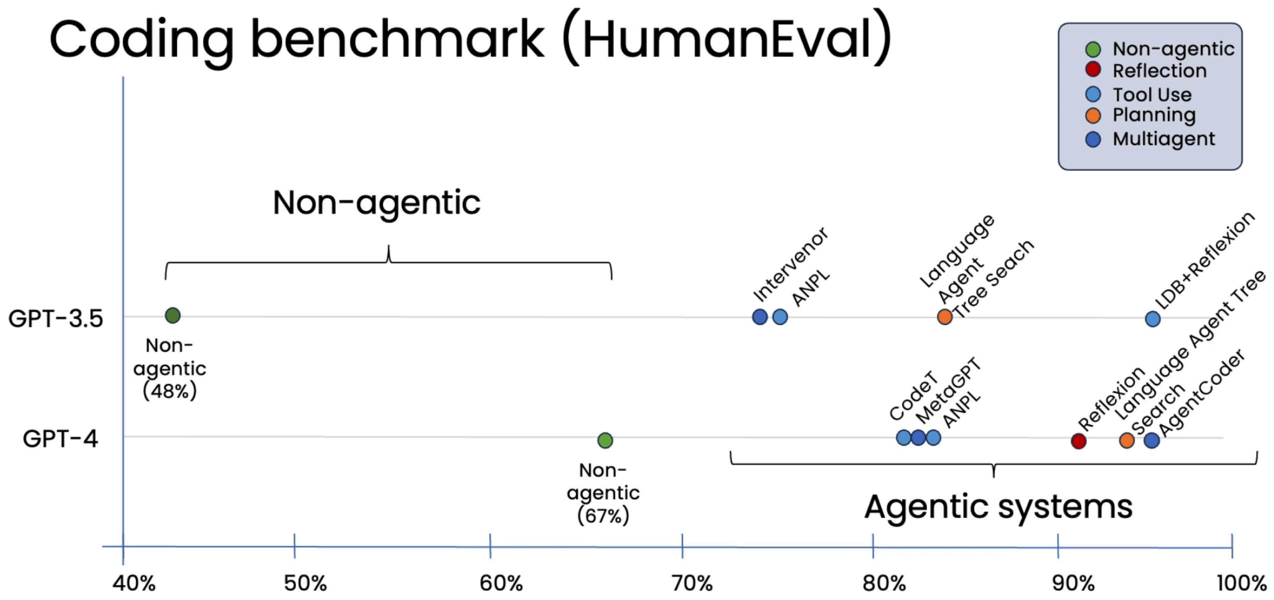


red box = input

AI agentic can be less or more autonomous.

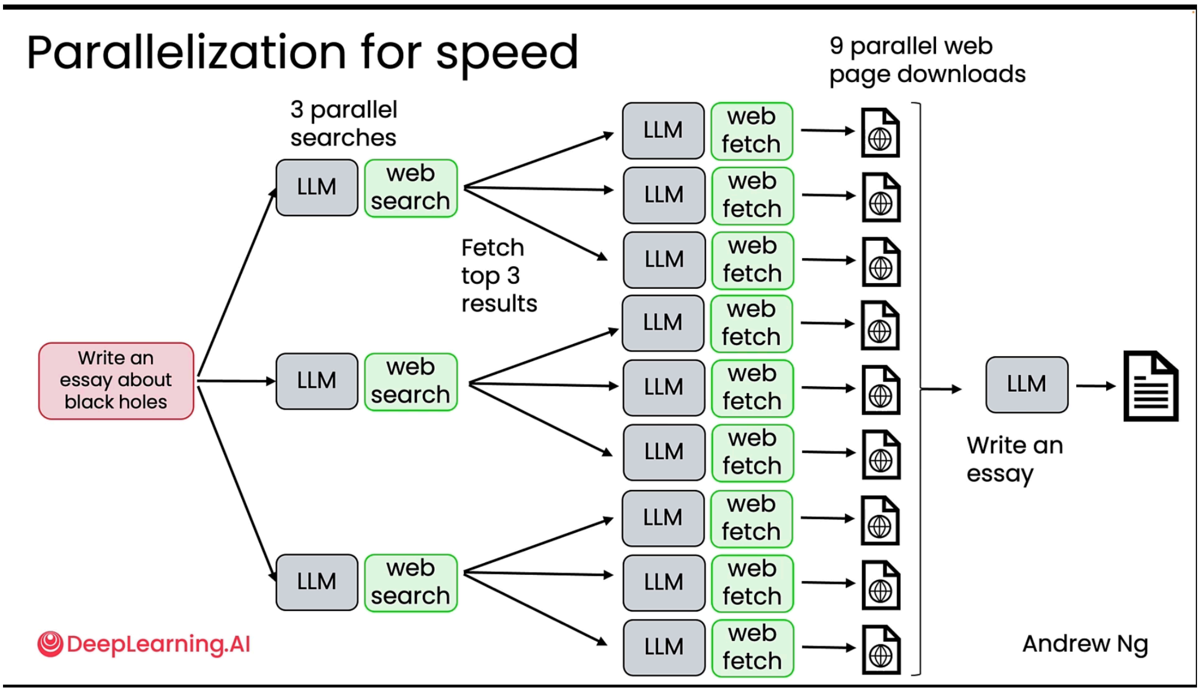


Comparison

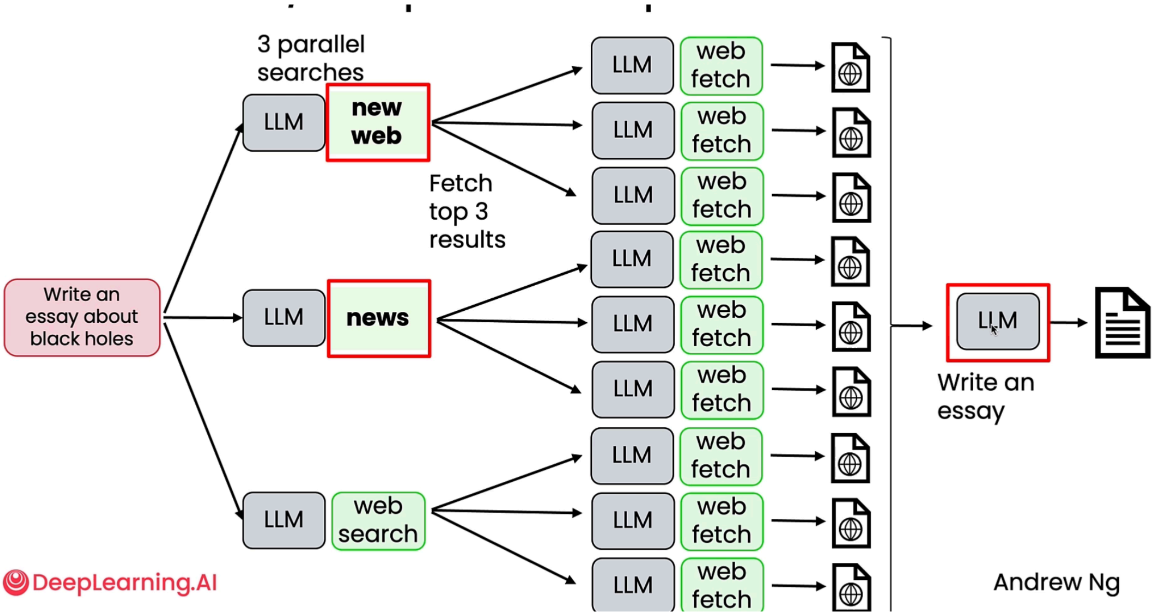


Benefit

- Allow you to do tasks that are previously not possible
- Parallelism: do certain things faster,
    - Example:



- Let us combine(swap/add) many other tools to output the desired outcome
  - We can also swap/add our component as well



Challenge

- Input: Do you have any black jeans or blue jeans?
  1. Check the inventory for the black jeans
  2. Check the inventory for the blue jeans
  3. Respond to the customer
- Input: I would like to return the beach towel I bought
  1. Verify customer purchase
  2. Check return policy
  3. if return allowed = "yes", then:

     a. issues return packing slip

     b. Set the database record to "return pending."

Task Decomposition: Identifying the steps in a workflow

Direct generation:	3-step workflow:	5-step workflow:
1. Write an essay on topic X	1. Write an essay outline on topic X	1. Write an essay outline on topic X
	2. Search web	2. Search web
	3. Write the essay	3. Write the first draft
Not good enough...	Still not good enough...	4. Consider what parts need revision
		5. Revise your draft

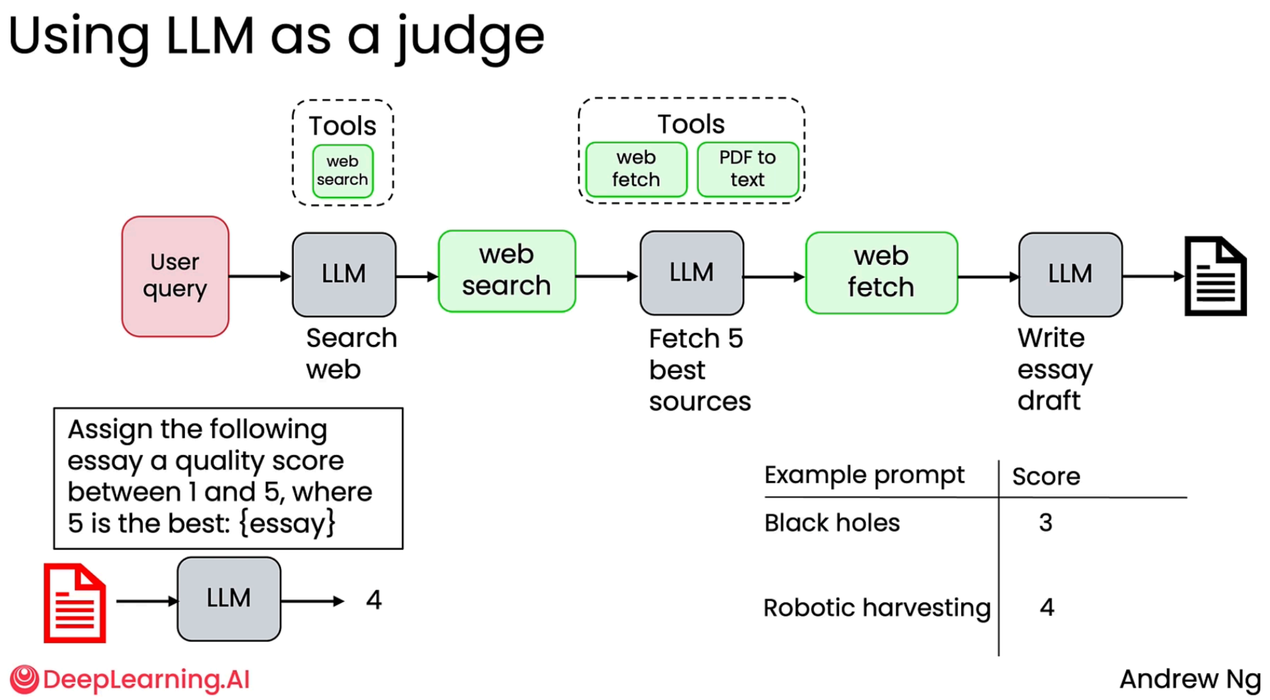
Building Blocks

1. Models:
   1. LLMs: text-generation, tool use, information extraction
   2. AI models: PDF-to-text, text-to-speech, image analysis....
2. Tools:
   1. API: web search, get real-time data, send email, check calendar,...
   2. Information retrieval: RAG
   3. Code execution: Basic calculator, data analysis\

Evaluation

We evaluate the agent by comparing the output with our desired outcome. Is it satisfactory? If not, then improve the agent

We can also use other LLMs to judge the output, too



all in all:

- can evaluate using code (objective evals) or use LLMs as judges (subjective evals)
- Two types of evals: End-to-end and component-level
- Examine traces to perform error analysis

Agentic Design Pattern

This allows the AI model to go beyond the single question-and-answer interaction. Instead of just responding once, the AI can reason, use tools, plan, and even collaborate with other AI agents to solve complex problems.

Step

1. **Reflection:** review and critique its own output
    1. Identify output
    2. iterate to improve
2. **Tool use:** web search, code exe, API
3. **Planning:** decide what order to do them and adapt if something goes wrong
4. **Multi-agent collaboration:** multi agent work together, and each has its own specialization (refer to the image)

Multi-agentic workflows

Multiagent Debate

Task	Single agent	Multi-agent
Biographies	66.0%	**73.8%**
MMLU	63.9%	**71.1%**
Chess move	29.3%	**45.2%**

(Du et al., 2023)